

Computer Vision:

Week-5 (16-20 Feb) Theory Lectures

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Review

- Intro to Feature Matching
- Feature detection Vs Descriptors
- Brute-Force & FLANN Feature Matching
- Ratio Test
- Numericals

Classical Segmentation and Shape Analysis

In **Computer Vision**, Classical Segmentation and Shape Analysis is the process of partitioning an image into meaningful "blobs" and then using geometry to describe what those blobs are.

Think of it as the transition from seeing pixels to recognizing objects.

Classical Segmentation (The "Isolation" Phase)

Segmentation is the process of grouping pixels together based on common characteristics like color, intensity, or texture.

Classical Segmentation (The "Isolation" Phase)

- **Thresholding:** The simplest form of segmentation. It turns a grayscale image into a binary (black and white) image by picking a "cutoff" value.
 - . *Example:* If a pixel is darker than 127, make it black; otherwise, make it white.
- **Edge-Based Segmentation:** Finding the boundaries where color or brightness changes abruptly. Since these edges are often "broken" lines, we use **Contour Tracing** to connect them into a solid outline.
- **Region-Based Segmentation:** Starting with a "seed" pixel and "growing" the region by adding neighboring pixels that look similar.

Connected Components Labeling (CCL):

Once you have a binary image, CCL scans the pixels and gives every separate "blob" a unique ID number so the computer can treat them as individual objects.

2. Shape Analysis (The "Description" Phase)

- Once an object is isolated, we need to describe it using numbers (descriptors).
- This allows a computer to distinguish a "square" from a "circle."

Basic Geometric Descriptors

Area & Perimeter: Counting the total pixels inside a shape and the number of pixels on its boundary.

Centroid: The geometric center (the "average" position) of all pixels in the shape.

Aspect Ratio: The ratio of the width to the height of the object's bounding box.

Example: A standing person has a high aspect ratio; a car has a low one.

Advanced Structural Descriptors

Circularity: A formula $(4\pi \times \frac{Area}{Perimeter^2})$

that tells us how "round" an object is.

A perfect circle equals 1.0.

Advanced Structural Descriptors

Convex Hull:

- Imagine wrapping a rubber band around an object.
- The shape the rubber band takes is the Convex Hull.



Solidity:

The ratio of the object's area to its Convex Hull area. A "star" shape has low solidity because of the empty space between its points.

Advanced Structural Descriptors

Hu Moments:

A set of seven numbers that describe a shape's "fingerprint." These numbers stay the same even if the object is **rotated, resized, or moved** in the image.

Advanced Structural Descriptors

<i>Step</i>	<i>Action</i>	<i>Tool Used</i>
<i>1. Isolate</i>	Separate object from	Thresholding / Canny
	background	Edges
<i>2. Group</i>	Identify individual "blobs"	Connected Components
<i>3. Measure</i>	Calculate size and	Area, Perimeter, Bounding
	orientation	Box
<i>4. Identify</i>	Determine "What is this?"	Hu Moments, Circularity

Applications

1. Industrial Quality Control (The "Sorter")

In a factory, a camera hangs over a conveyor belt. It must decide in milliseconds if a part is defective.

- **Technique: Thresholding + Shape Descriptors.**
- **How it works:**
 1. The image is converted to binary to isolate the metallic part from the black belt. .
 2. **Connected Components** identify each individual bolt or gear.
 3. The computer calculates the **Solidity** and **Circularity**.

-Example: If a "circular" gear has a tooth missing, its Solidity will drop below a specific threshold (e.g., 0.95), and a robotic arm will flick it into the "Reject" bin.

Applications

2. Medical Imaging (The "Pathologist")

Before a doctor looks at an MRI or a blood smear, classical algorithms "pre-process" the data to highlight areas of interest.

- **Technique: Active Contours (Snakes) + Region Growing.**
- **How it works:**
 - 1. A doctor clicks near a potential tumor (the "seed").
 - 2. **Region Growing** expands the selection to all nearby pixels with similar density.
 - 3. **Active Contours** then act like a "shrink-wrap," tightening around the exact boundary of the tumor.
- **Example:** Calculating the **Volume** of a tumor over six months to see if a treatment is shrinking it.

Calculations

Problem 1: The Circularity of a Square

Scenario: You have a perfectly square object in an image with a side length of 10 pixels. Calculate its circularity to determine how far it "deviates" from a perfect circle.

$$\textit{Side } (s) = 10 \textit{ pixels}$$

Calculations

Formulas :

$$Area(A) = s^2$$

$$Perimeter(P) = 4s$$

$$Circularity(C) = \frac{4\pi A}{P^2}$$

Calculations

Solution :

CalculateArea : $A = 10^2 = 100 \text{ pixels}$.

CalculatePerimeter : $P = 4 \times 10 = 40 \text{ pixels}$.

CalculateCircularity :

Calculations

$$C = \frac{4 \times \pi \times 100}{40^2}$$

$$C = \frac{400\pi}{1600} = \frac{\pi}{4}$$

$$C \approx \frac{3.14159}{4} \approx 0.785$$

A square always has a circularity of approximately **0.785**.

(A perfect circle is 1.0)

Problem 2: Aspect Ratio and Extent

Scenario:

A security camera detects a "blob" that has an area of 1,200 pixels. The bounding box around this blob has a width of 30 pixels and a height of 80 pixels. Determine if this is likely a standing person or a car.

Problem 2: Aspect Ratio and Extent

$$Area_{object} = 1200$$

$$Width(W) = 30$$

$$Height(H) = 80$$

Formulas :

$$AspectRatio = \frac{Width}{Height}$$

$$Extent = \frac{Area_{object}}{Area_{BoundingBox}}$$

Problem 2: Aspect Ratio and Extent

Solution :

$$\text{Aspect Ratio : } AR = \frac{30}{80} = 0.375$$

$$\text{Bounding BoxArea : } A_{bbox} = 30 \times 80 = 2400$$

$$\text{Extent : } E = \frac{1200}{2400} = 0.5$$

Practice Problem 1: Circularity of a Rectangle

Scenario: An industrial camera identifies a metallic bar. The bar is a rectangle with a **Width of 20 pixels** and a **Height of 5 pixels**.

***Your Task:** 1. Calculate the Area (A).*

2. Calculate the Perimeter (P).

3. Calculate the Circularity (C).